

Pristinex[®] TABLETS

Ketoconazole 200mg

Composition : Each tablet contains Ketoconazole 200 mg.

Properties : Ketoconazole is a imidazole antifungal agent that impairs the synthesis of ergosterol, which is a vital component of fungal cell membrane and therefore alters the permeability of the cell membrane of the following common dermatophytes and yeast. Dermatophytes: *Trichophyton rubrum*, *T. mentagrophytes*, *T. tonsurans*, *Microsporum canis*, *M. audouinii*, *M. gypseum* and *Epidermophyton floccosum*; Yeast: *Candida albicans*, *Malassezia ovale* (*Pityrosporum ovale*) and *C. tropicalis*; and the common organism responsible for tinea versicolor, *Malassezia furfur* (*Pityrosporum orbiculare*).

Pharmacology : The absorption of ketoconazole from the gastro-intestinal tract is variable and increases with decreasing stomach pH. It is recommended that ketoconazole is given with food to increase absorption and reduce gastro-intestinal disturbances, although reduced rate and extent of absorption have occurred when given after a meal. Mean peak plasma concentrations of about 3.5 µg per mL have been obtained 2 hours after administration of 200 mg by mouth. Systemic absorption following topical or vaginal application in healthy subjects is minimal. Ketoconazole is more than 90% bound to plasma proteins, mainly albumin. It is widely distributed but penetration into the cerebrospinal fluid is poor. The elimination of ketoconazole is reported to be biphasic, with an initial half-life of 2 hours and a terminal half-life of about 8 hours. Ketoconazole is metabolised in the liver to inactive metabolites. It is excreted as metabolites and unchanged drug chiefly in the faeces; some is excreted in the urine.

Indications :

- Infections of the skin, hair and nails induced by dermatophytes and/or yeasts (dermatophytosis, onychomycosis, *Candida* perionyxis, pityriasis versicolor, pityriasis capitis, pityrosporum folliculitis, chronic mucocutaneous candidosis), provided these infections cannot be treated topically because of the site or extension of the lesion or deep affection of the skin, or if they fail to respond to local therapy.
 - Yeast infection of the gastrointestinal tract.
 - Chronic, recurrent vaginal candidosis provided this infection fails to respond to local therapy.
 - Systemic fungal infections such as systemic candidosis, paracoccidioidomycosis, histoplasmosis, coccidioidomycosis, blastomycosis.
 - Prophylactic treatment of patients with decreased defense mechanisms (inherited, caused by illness or drugs) involving an increased risk of fungal infections.
- Ketoconazole does not penetrate well in the CNS. Therefore, fungal meningitis should not be treated with oral ketoconazole.

Dosage and Administration : Ketoconazole should be taken during meals for maximal absorption. Infections of the skin, hair, and mucosa induced by dermatophytes and/or yeasts, and systemic infections, that cannot be treated topically because of the site or the extent of the lesion or deep infection of the skin:

Adults

One tablet (200 mg) once daily with a meal. When no adequate response is obtained with this dose, the dose should be increased to two tablets (400 mg) once daily. Adults with vaginal candidosis: Two tablets (400 mg) once daily with a meal.

Children

- Children weighing from 15 to 30 kg: ½ tablet (= 100 mg) once daily with a meal.
- Children weighing more than 30 kg: Same as for adults.

The usual duration of treatment is:

- Vaginal candidosis: 5 consecutive days
- Skin mycosis induced by dermatophytes: approximately 4 weeks
- Pityriasis versicolor: 10 days
- Oral and skin mycosis induced by *Candida*: 2-3 weeks
- Hair infections: 1-2 months
- Paracoccidioidomycosis, histoplasmosis, coccidioidomycosis: The usual duration of therapy is 6 months.

For all indications, treatment should be continued without interruption until clinical parameters or laboratory tests indicate that the fungal infection has resolved. An inadequate treatment period may lead to recurrence of the active infection. However, treatment should be stopped immediately and liver function testing should be conducted when signs and symptoms suggestive of hepatitis such as anorexia, nausea, vomiting, fatigue, jaundice, abdominal pain or dark urine occur.

Special Patient Population: Hepatic Impairment (see Contraindications).

Side Effects/Adverse Reactions : Gastro-intestinal disturbances are the most frequently reported adverse effect following the oral administration of ketoconazole. Nausea and vomiting have been reported in 3 to 10% of patients, and diarrhoea, constipation, and abdominal pain in about 1%. These adverse effects are dose-related and may be minimised by giving ketoconazole with food. Ketoconazole interferes with steroid biosynthesis and reported endocrine effects include gynaecomastia and adrenal cortex suppression.

Other adverse effects include hypersensitivity reactions, pruritus, rash, headache, dizziness, and somnolence. Thrombocytopenia, paraesthesia, and photophobia have been reported rarely.

Asymptomatic, transient elevations in serum concentrations of liver enzymes may occur.

Post Marketing Experience

Hepato-biliary Disorders

Very rare: serious hepatotoxicity, including hepatitis cholestatic, biopsy-confirmed hepatic necrosis, cirrhosis, hepatic failure including cases resulting in transplantation or death (see Precautions/Warnings).

Precautions / Warning : In normal patients, liver function tests should be performed before commencement of long-term treatment with ketoconazole and then at least monthly throughout treatment.

Do not use on children under 2 years of age except under the advice and supervision of a doctor.

Because of the risk for serious hepatotoxicity, Pristinex should be used only when the potential benefits are considered to outweigh the potential risks, taking into consideration the availability of other effective antifungal therapy. Assess liver function, prior to treatment to rule out acute or chronic liver disease, and monitor at frequent and regular intervals during treatment, and at the first signs or symptoms of possible hepatotoxicity.

Hepatotoxicity

Very rare case of serious hepatotoxicity, including cases with a fatal outcome or requiring liver transplantation have occurred with the use of oral ketoconazole. Some patients had no obvious risk factors for liver disease. Cases have been reported that occurred within the first month of treatment, including some within the first week.

The cumulative dose of the treatment is a risk factor for serious hepatotoxicity. Factors which may increase the risk of hepatitis are prolonged treatment with ketoconazole tablets, females over 50 years of age, previous treatment with griseofulvin, a history of liver disease, known drug intolerance and concurrent use of medication which compromises liver function. A period of one month should be allowed between cessation of griseofulvin treatment and commencement treatment with ketoconazole tablets because of an apparent association between recent griseofulvin therapy and hepatic reactions to ketoconazole tablets.

Monitor liver function in all patients receiving treatment with ketoconazole tablets (see Monitoring of hepatic function).

Patients should be instructed to promptly report to their physician signs and symptoms suggestive of hepatitis such as anorexia, nausea, vomiting, fatigue, jaundice, abdominal pain or dark urine. In these patients, treatment should be stopped immediately and liver function should be conducted.

Monitoring of hepatic function

Monitor liver function in all patients receiving treatment with ketoconazole tablets. Monitor liver function prior to treatment to rule out acute or chronic liver disease (see Contraindications), after two weeks of treatment and then on a monthly basis and at the first sign or symptoms of possible hepatic toxicity. When the liver function tests indicate liver injury, the treatment should be stopped immediately.

A risk and benefit evaluation should be made before oral ketoconazole is used in cases of non-life threatening diseases requiring long treatment periods.

In patients with elevated liver enzymes, or who have experienced liver toxicity with other drugs, treatment should not be started unless the expected benefit exceeds the risk of hepatic injury. In such cases, close monitoring of the liver enzymes is necessary.

Contraindications : Ketoconazole should not be administered to patients with acute or chronic liver disease or with a known hypersensitivity to the drug. Co-administration of terfenadine or astemizole with ketoconazole tablets is contraindicated. Avoid the use of ketoconazole tablets with cisapride.

Interactions : Concomitant administration of drugs that reduce stomach acidity, such as antimuscarinic agents, antacids, and H₂-receptor antagonists, may reduce the absorption of ketoconazole.

Concomitant administration of ketoconazole with rifampicin results in decreased antifungal activity; the addition of isoniazid leads to further decrease. Concurrent use of ketoconazole and phenytoin may reduce serum-ketoconazole concentrations, probably due to induction of hepatic enzymes.

Ketoconazole inhibits certain hepatic oxidase enzymes and may account for increases in plasma concentrations of some hepatically metabolised drugs such as astemizole, cisapride, corticosteroids, cyclosporin, oral anticoagulants, and terfenadine. A disulfiram-like reaction may occur in patients taking ketoconazole after drinking alcohol. The efficacy of oral contraceptives may be reduced. Anticoagulant effect of coumarin-like drugs.

Pregnancy and Lactation

Ketoconazole has been shown to be teratogenic in animal studies and its use is generally not recommended during pregnancy. Avoid use in porphyria and in nursing mother.

Overdosage and Treatment : In the event of accidental overdosage, symptoms such as abnormal liver functions, dizziness or other adverse effects as mentioned may be experienced.

Treatment for overdosage consists of supportive measures. Within the first hour after ingestion, gastric lavage may be performed. Activated charcoal may be given if considered appropriate.

Effects on Ability to Drive and Use Machines : No effects have been observed.

Storage : Store below 30°C. Protect from light.

**KEEP ALL MEDICINES OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK**

Shelf-life : The expiry date is indicated on the packaging.

Presentation : White, plain round tablets with one breakline on one side, 10mm in diameter. 100's in blister packs of 10's.

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